Relational Schema — E-Commerce Backend

Derived directly from the ER diagram. Notation: `PK` = primary key, `FK` = foreign key.

Staff

Column	Type	Constraint
staff_id	INTEGER	PK
name	TEXT	NOT NULL
role	TEXT	NOT NULL

`Staff(staff_id PK, name, role)`

Customer

Column	Type	Constraint
customer_id	INTEGER	PK
name	TEXT	NOT NULL
email	TEXT	NOT NULL, UNIQUE
phone	TEXT	

`Customer(customer_id PK, name, email, phone)`

Product

Column	Type	Constraint
product_id	INTEGER	PK
name	TEXT	NOT NULL
price	DECIMAL(10,2)	NOT NULL

Column	Type	Constraint
quantity_in_stock	INTEGER	NOT NULL
added_by_staff_id	INTEGER	FK → Staff(staff_id)

`Product(product_id PK, name, price, quantity_in_stock, added_by_staff_id FK → Staff)`

CreditCard

Column	Type	Constraint
card_id	INTEGER	PK
customer_id	INTEGER	FK → Customer(customer_id)
card_last4	TEXT	NOT NULL
expiry	TEXT	NOT NULL
billing_zip	TEXT	

`CreditCard(card_id PK, customer_id FK → Customer, card_last4, expiry, billing_zip)`

Purchase

Column	Type	Constraint
purchase_id	INTEGER	PK
customer_id	INTEGER	FK → Customer(customer_id)
product_id	INTEGER	FK → Product(product_id)
card_id	INTEGER	FK → CreditCard(card_id)
quantity	INTEGER	NOT NULL
purchase_date	TEXT (ISO date)	NOT NULL

`Purchase(purchase_id PK, customer_id FK → Customer, product_id FK → Product, card_id FK → CreditCard, quantity, purchase_date)`

Summary

5 relations total (exceeds the individual-project minimum of 4): **Staff, Customer, Product, CreditCard, Purchase**. Every foreign key references a primary key in another relation, matching the cardinalities documented in the ER diagram. The full DDL is in schema.sql.